

1. USER STORIES AND ACCEPTANCE CRITERIA

MODULE 1: AUTHENTICATION & AUTHORIZATION

US-AUTH-001: User Registration

User Story: As a new student, I want to create an account on StudyFlow using my email and personal details so that I can manage my academic tasks.

- **AC-AUTH-01.1 (Valid Registration):**
 - **Given** the user is on the registration page (/register),
 - **When** they enter a valid first name, last name, unique email address, and a strong password (minimum 8 characters),
 - **And** click the "Register" button,
 - **Then** the account should be created in the database,
 - **And** a JWT authentication token should be generated and stored in local storage,
 - **And** the user should be redirected to the Dashboard (/dashboard).
- **AC-AUTH-01.2 (Duplicate Email Validation):**
 - **Given** a user with the email student@studyflow.com already exists,
 - **When** a new user attempts to register with student@studyflow.com,
 - **Then** the system should reject the request with HTTP Status 400 Bad Request,
 - **And** display an inline error message: "Email is already registered."

US-AUTH-002: User Login

User Story: As a registered student, I want to log in using my email and password so that I can access my personalized dashboard.

- **AC-AUTH-02.1 (Successful Login):**
 - **Given** the user is on the login page (/login),
 - **When** they provide valid account credentials,
 - **And** click "Login",
 - **Then** HTTP Status 200 OK should be returned,
 - **And** the user session should be established, redirecting them to /dashboard.

- **AC-AUTH-02.2 (Invalid Credentials):**
 - **Given** the user inputs an incorrect password or unregistered email,
 - **When** they submit the login form,
 - **Then** HTTP Status 401 Unauthorized should be returned,
 - **And** an error message "Invalid email or password" should be displayed.

US-AUTH-003: User Logout

User Story: As a logged-in student, I want to log out of my account so that I can protect my personal data on shared devices.

- **AC-AUTH-03.1 (Successful Logout):**
 - **Given** the user is logged into the application,
 - **When** they click the "Logout" button in the Sidebar,
 - **Then** the authentication token should be removed from localStorage,
 - **And** the user should be redirected to the Login page (/login).

MODULE 2: USER PROFILE & NAVIGATION

US-PROF-001: Sidebar User Information Display

User Story: As an authenticated user, I want to see my actual full name, dynamic initials, and profile link in the Sidebar so that I can verify my active session.

- **AC-PROF-01.1 (Dynamic Profile Info):**
 - **Given** a user is logged in as "John Doe",
 - **When** the Sidebar component renders,
 - **Then** the name "John Doe" and initials "JD" should be displayed dynamically from session data,
 - **And** no hardcoded default user data should appear.
- **AC-PROF-01.2 (Profile Navigation):**
 - **Given** the user clicks on their profile avatar/name block in the Sidebar,
 - **When** the click event triggers,
 - **Then** the application should navigate to the User Profile page (/profile).

MODULE 3: COURSE MANAGEMENT

US-CRS-001: Course Creation and Deletion

User Story: As a student, I want to create and delete academic courses so that I can keep my course list relevant to my current semester.

- **AC-CRS-01.1 (Create Course):**
 - **Given** the user opens the Add Course modal,
 - **When** they enter a course title (e.g., "Software Quality Assurance") and submit,
 - **Then** the new course should be saved via POST `/api/courses` and rendered in the courses list.
- **AC-CRS-01.2 (Delete Course):**
 - **Given** an existing course card on the Courses page,
 - **When** the user clicks the "Delete Course" button,
 - **Then** a confirmation modal should prompt the user,
 - **And** upon confirmation, DELETE `/api/courses/:id` should execute, removing the course from the UI.

MODULE 4: TASK & SCHEDULE MANAGEMENT

US-TSK-001: Task Creation and Status Tracking

User Story: As a student, I want to create tasks associated with specific courses and mark them as complete so that I can track my academic progress.

- **AC-TSK-01.1 (Create Task with Deadline):**
 - **Given** the user is on the Tasks page,
 - **When** they enter a task title, select a course, and set a due date,
 - **Then** the task should be saved with status "Pending" and appear in chronological order.
- **AC-TSK-01.2 (Complete Task):**
 - **Given** a pending task item in the list,
 - **When** the user checks the completion checkbox,
 - **Then** the status should update to "Completed" and the active task counter should decrement.

MODULE 5: EMAIL NOTIFICATIONS & PASSWORD RECOVERY

US-NTF-001: Automated Email Notifications

User Story: As a user, I want to receive email notifications upon registration and password reset requests so that I can manage my account security.

- **AC-NTF-01.1 (Welcome Email Delivery):**
 - **Given** a new user completes registration,
 - **When** the account is successfully created,
 - **Then** an automated welcome email should be dispatched via SMTP within 10 seconds.
- **AC-NTF-01.2 (Forgot Password Email):**
 - **Given** the user submits their registered email address on the /forgot - password page,
 - **When** the request is submitted,
 - **Then** a password reset link containing a secure token should be emailed to the user.

2. REQUIREMENTS TRACEABILITY MATRIX (RTM)

Requirement ID	Requirement Description	Acceptance Criteria ID	Test Case ID	Current Status	Linked Defect ID
REQ-AUTH-01	User Registration with JWT	AC-AUTH-01.1, AC-AUTH-01.2	TC-AUTH-001, TC-AUTH-002	PASS	N/A
REQ-AUTH-02	User Authentication (Login)	AC-AUTH-02.1, AC-AUTH-02.2	TC-AUTH-005, TC-AUTH-006	PASS	N/A
REQ-AUTH-03	Logout & Token Invalidation	AC-AUTH-03.1	TC-AUTH-008	FAIL	BUG-004

REQ-PROF-01	Dynamic User Data in Sidebar	AC-PROF-01.1	TC-DASH-012	FAIL	BUG-004
REQ-PROF-02	Profile Navigation from Sidebar	AC-PROF-01.2	TC-DASH-013	FAIL	BUG-004
REQ-CRS-01	Course Creation (Create CRUD)	AC-CRS-01.1	TC-CRS-021	PASS	N/A
REQ-CRS-02	Course Deletion (Delete CRUD)	AC-CRS-01.2	TC-CRS-023	FAIL	BUG-003
REQ-TSK-01	Task Creation and Deadlines	AC-TSK-01.1	TC-TSK-031	PASS	N/A
REQ-TSK-02	Task Completion Toggle	AC-TSK-01.2	TC-TSK-032	PASS	N/A
REQ-NTF-01	Welcome Email Notification	AC-NTF-01.1	TC-NTF-045	FAIL	BUG-005
REQ-NTF-02	Password Reset Instructions Email	AC-NTF-01.2	TC-NTF-046	FAIL	BUG-005
REQ-UI-01	Mobile Responsive Layout Adaptability	N/A (Non-Functional)	TC-DASH-014, TC-DASH-020	FAIL	BUG-001

REQ-UI-02	New User Empty State Initialization	N/A (Non- Functional)	TC-DASH- 011	FAIL	BUG- 002
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